Tutorial 3 Mini-Lecture: Atmospheres: Structure, Clouds, & Climate

The concepts behind Worksheet 3 • Lectures 5–6

Kapteyn Astronomical Institute
University of Groningen

2026

Jupiter’s atmosphere. Credit: NASA/JPL-Caltech/SwRI/MSSS

What Worksheet 3 trains

Five problems, all built on Lectures 5–6. The physics you need, per problem:

1. **How thick is an atmosphere?**: scale height, hydrostatic balance, column mass.
2. **Sunlight in, infrared out**: equilibrium temperature, the one-layer greenhouse.
3. **From vapour to cloud base**: Clausius-Clapeyron, dew point, the LCL, one sketch.
4. **Winds on a rotating planet**: Coriolis parameter, Rossby number, geostrophic balance.
5. **Climate on the edge**: the faint young Sun, ice-albedo bistability, the carbonate-silicate thermostat.

Every problem has the shape and level of one exam question. All parts are analytical; no computer is needed.

How thick is an atmosphere?

Two workhorses of Lecture 5:

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}} \qquad P_0 = m_{\mathrm{col}}\, g$$

- H compares thermal energy against gravitational confinement: Earth 8.4 km, Titan 20 km.
- Integrating $\mathrm{d}P = -\rho g\, \mathrm{d}z$ over the column: surface pressure is the weight of the air overhead.

Watch the ratio, not the terms

$H \propto T/g$: when two effects pull in opposite directions, put numbers on both ratios before deciding which one wins. Problem 1(b) turns exactly on this.

Ten tonnes of air stand on every square metre of Earth; the scale height says how thickly it is spread.

Sunlight in, infrared out

Balance absorbed sunlight against emitted infrared:

$$\boxed{(1 - A)\frac{S}{4} = \sigma T_{\mathrm{eq}}^4} \qquad T_s = 2^{1/4}\, T_{\mathrm{eq}} \quad (\varepsilon = 1)$$

- S falls with the inverse square of distance; A is the Bond albedo.
- Venus: brilliant clouds give $T_{\mathrm{eq}} = 227$ K, colder than Earth's 255 K, above a 737 K surface.

The greenhouse in one sentence

The atmosphere passes sunlight but blocks the surface's infrared, so the surface must heat up until its share leaks through.

The place to test any claim about the greenhouse budget is the top of the atmosphere: check whether absorbed sunlight and outgoing infrared still balance there.

From vapour to cloud base

The blackboard result of Lecture 6:

$$\boxed{P_{\mathrm{sat}}(T) = P_{\mathrm{ref}}\, e^{-\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)}}$$

- Invert it for the dew point T_d; rising air cools at $\Gamma_d = 9.8\ \mathrm{K\,km^{-1}}$ until it gets there.
- $z_{\mathrm{LCL}} = (T - T_d)/\Gamma_d$: the cloud base. Above it, the moist adiabat (~5 K km^{-1}) takes over.

The sketch part

Problem 3(c) asks for the T–z profile: two straight lines meeting at the LCL. Label both slopes; the kink is the physics.

One exponential governs every cloud in the solar system; L_v/R_v sets how sharply it bites.

Winds on a rotating planet

Three numbers decide the flow:

$$f = 2\Omega \sin\phi, \quad \mathrm{Ro} = \frac{U}{fL}, \quad u_g = \frac{1}{f\rho}\left|\frac{\partial P}{\partial y}\right|$$

- $\mathrm{Ro} \ll 1$: rotation wins, wind blows along isobars (geostrophic balance).
- $f = 0$ at the equator: no deflection, no cyclones there.

Slow vs fast rotators

The number of circulation cells follows from the strength of the Coriolis deflection. Reason it out from the Rossby number before you judge the claim in Problem 4(e).

Check the balance with numbers: a 2000 km low with 15 m s^{-1} winds has $\mathrm{Ro} \approx 0.06$; geostrophy predicts its wind to a few percent.

Climate on the edge

Long-term climate is a balance with feedbacks:

- Young Sun at 71%: T_{eff} = 234 K, yet the rocks record liquid water: the faint young Sun paradox.
- Ice-albedo feedback: two stable climates under the same Sun.
- Carbonate-silicate thermostat: warmer → faster weathering → less $\mathrm{CO_2}$ → cooler.

Reading the bistability figure

Classify each crossing from the local slopes: nudge the temperature slightly and ask whether absorption or emission grows faster. Problem 5(b) asks you to argue this from the curves.

A thermostat needs a working source and a working sink: Venus lost its water, Mars lost its volcanism, Earth kept both.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** ("show that …") to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Two parts read figures printed on the sheet (the saturation curves and the bistability diagram); argue from the curves, not from memory.

Worksheet and full solutions: `ips.formingworlds.space/worksheets.html`.
We discuss questions, not presentations of the solutions.